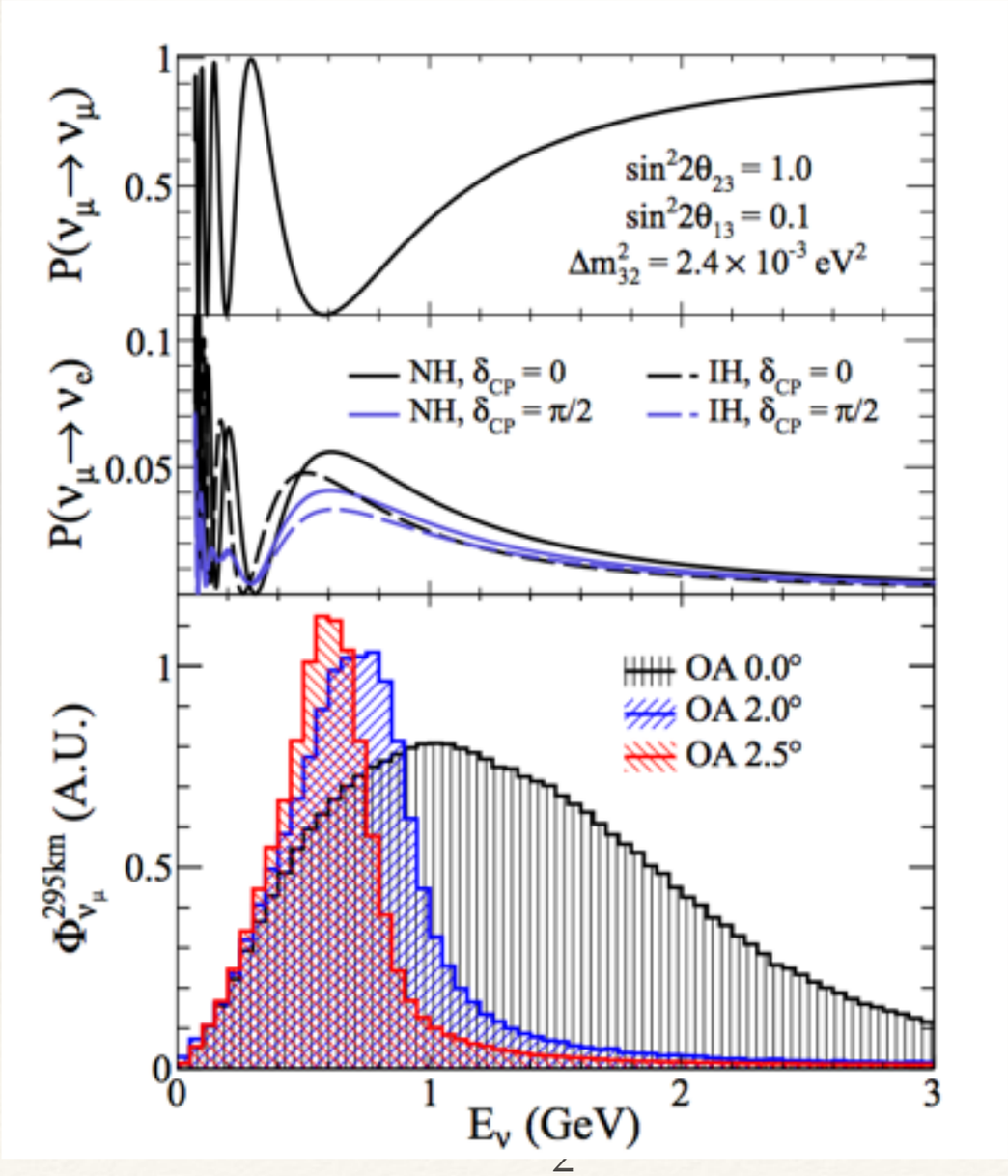
Optimizing the Off-axis Angle for T2K and T2HK

Mark Hartz - Kavli IPMU (WPI), University of Tokyo/TRIUMF

Off-axis Angle Choice

- The off-axis angle for T2K is set to 2.5 degrees
- This gives flux peaked near the first oscillation maximum and minimizes feed-down from NC backgrounds in the oscillation region



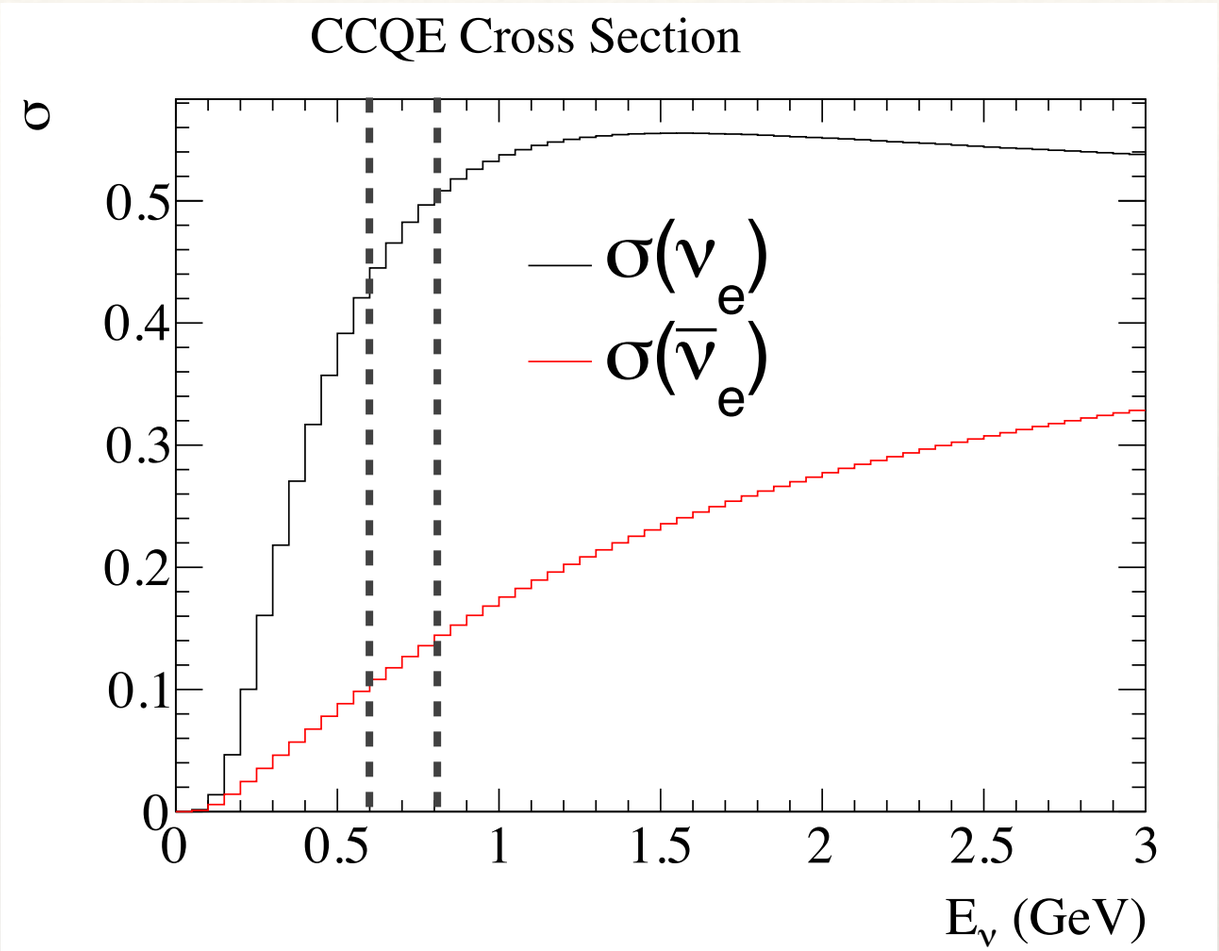
Off-axis Angle Optimization

Motivation for Re-optimization

- Since the T2K off-axis angle was chosen, there are new data that could change the choice of off-axis angle:
  - θ_{13} is relatively large, so the ν_e candidate rate is signal dominated
  - Improved reconstruction has significantly reduced the NCπ^0 background in ν_e reconstruction
  - CP violation measurements will use significant fractions of antineutrino enhanced data
- The beam-line at J-PARC can produce beams between 2 and 2.5 degrees off-axis
- We can check sensitivities with 2.0 and 2.25 degree off-axis fluxes against the nominal 2.5 degree flux. Is 2.5 degrees optimal?

Is Higher Energy Flux Better?

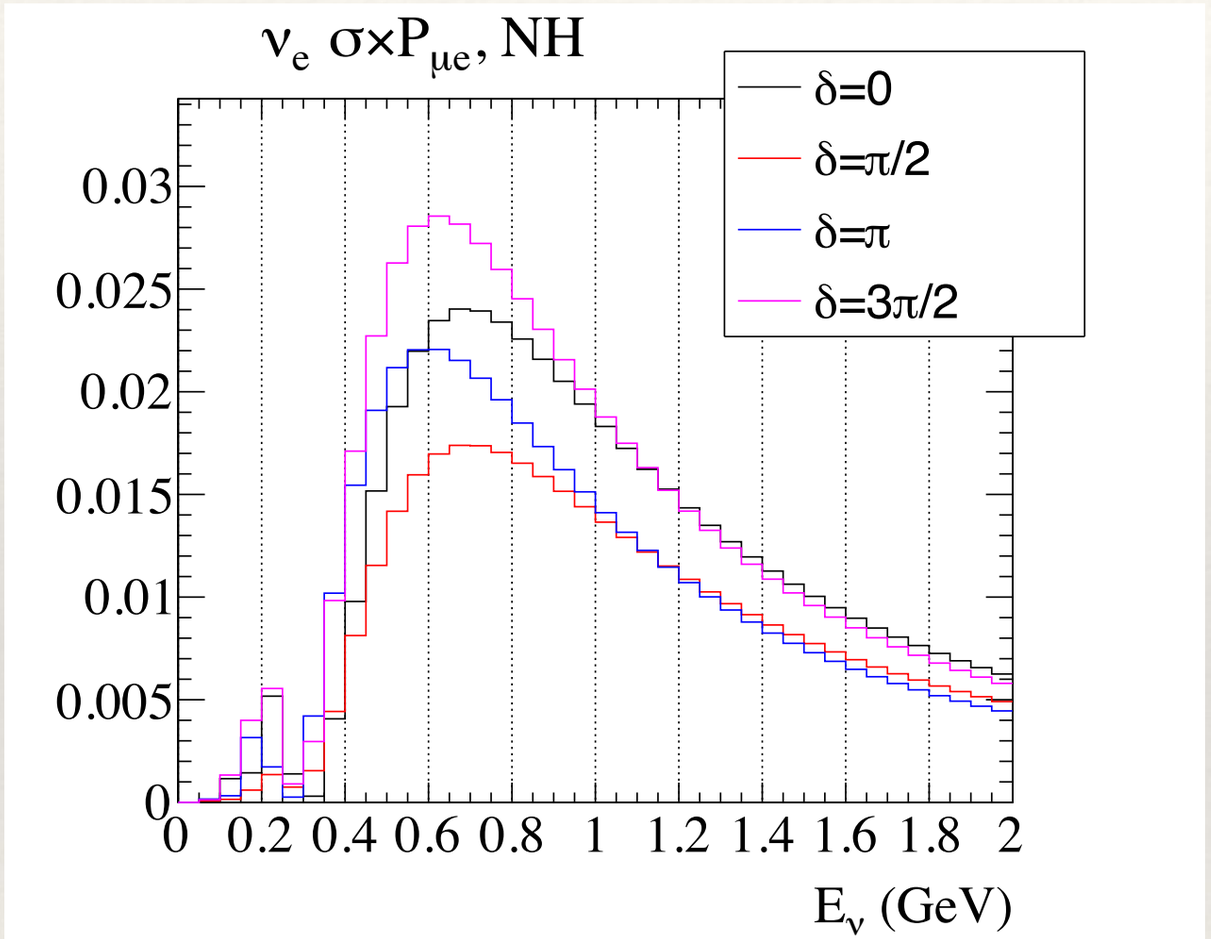
- The CCQE-like cross sections are rising in the <1 GeV region

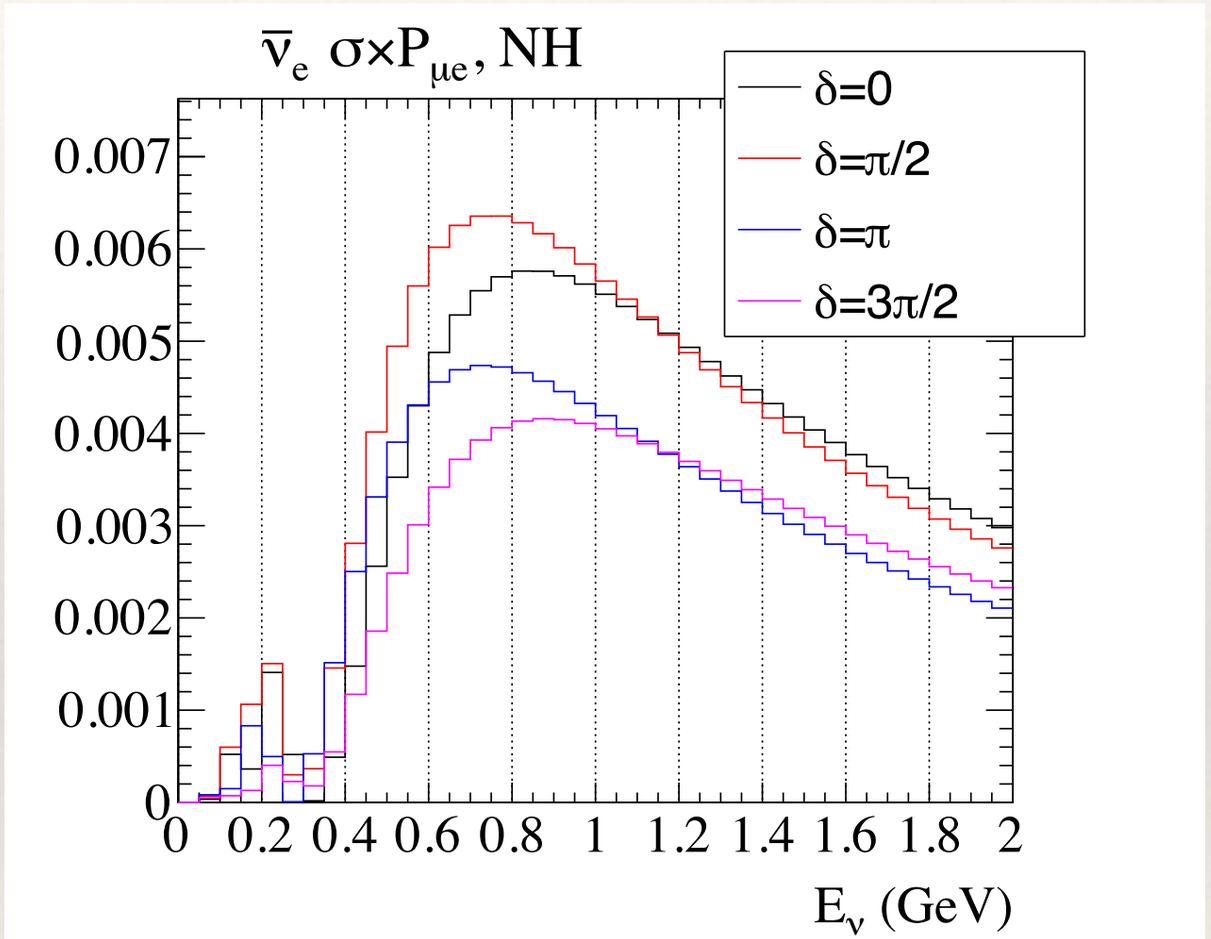


- Cross section at 800 MeV relative to 600 MeV:
  - neutrino: 1.21, antineutrino: 1.47
- Cross section gain could be even higher if 2-ring samples are used

Oscillation Probability (NH)

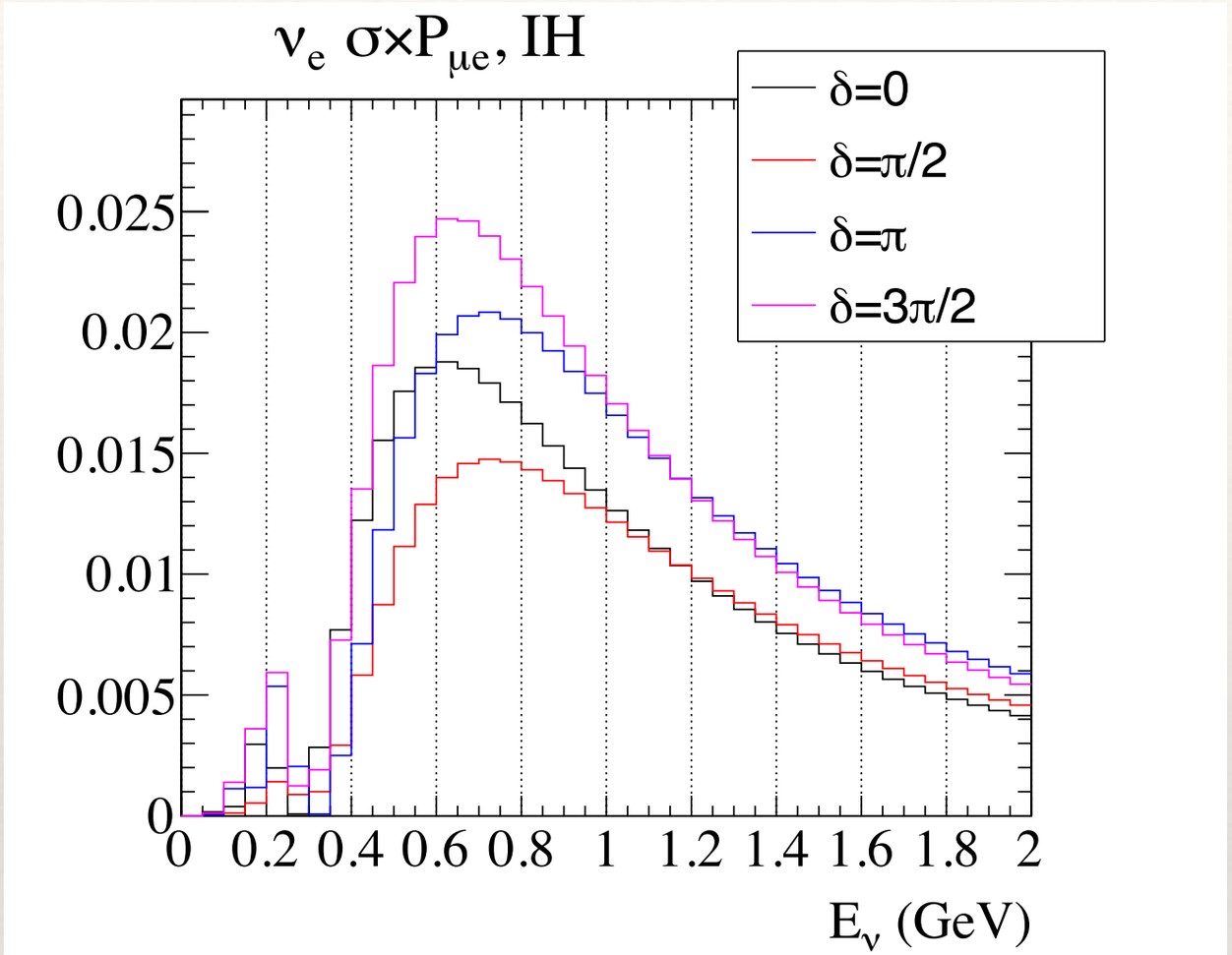
- To eyeball the optimal peak for the flux, we can look at the cross-section times oscillation probability:

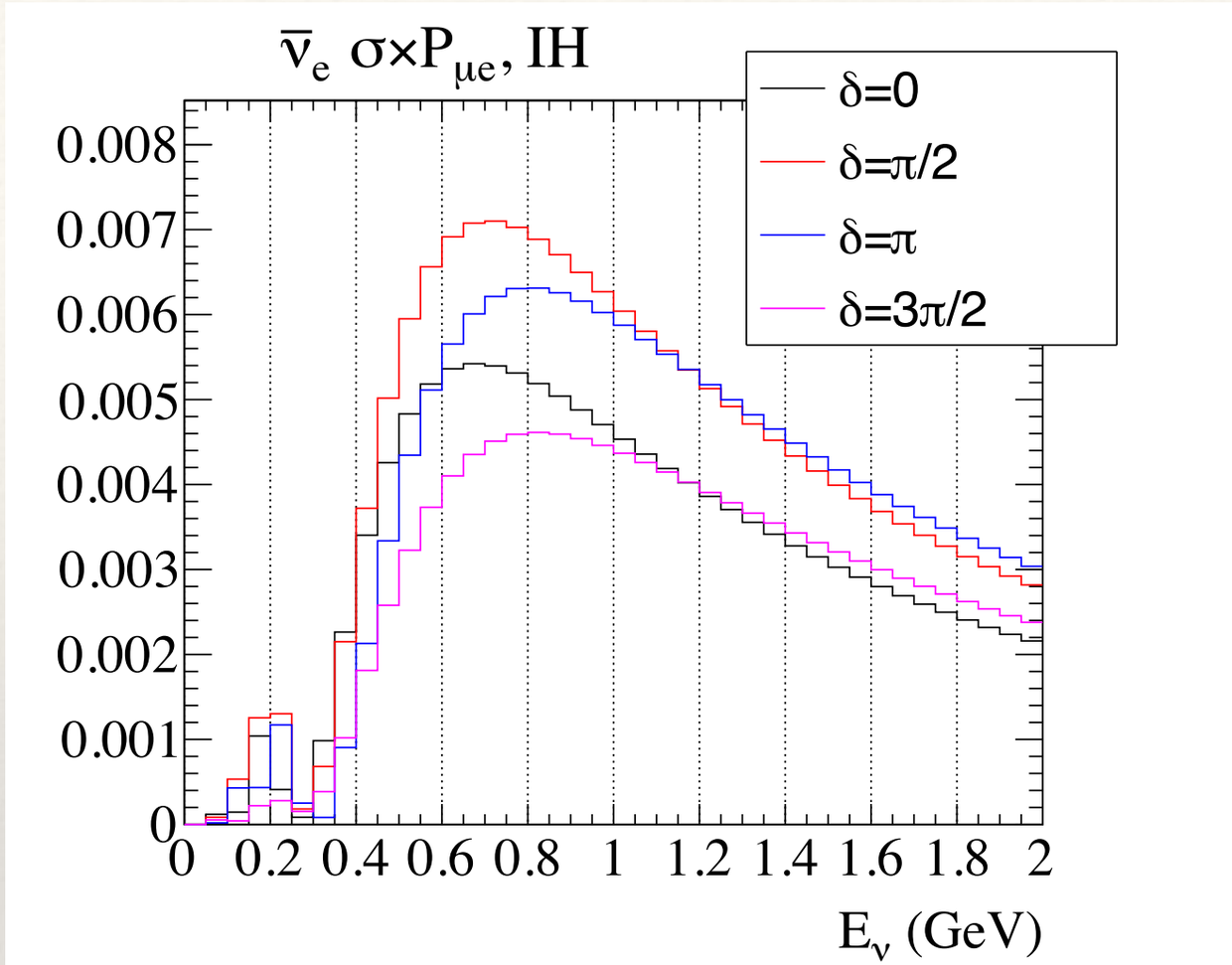




- All of the antineutrino curves are peaked at 0.7 GeV or greater. The neutrino curves are all peaked at 0.6 GeV or greater.

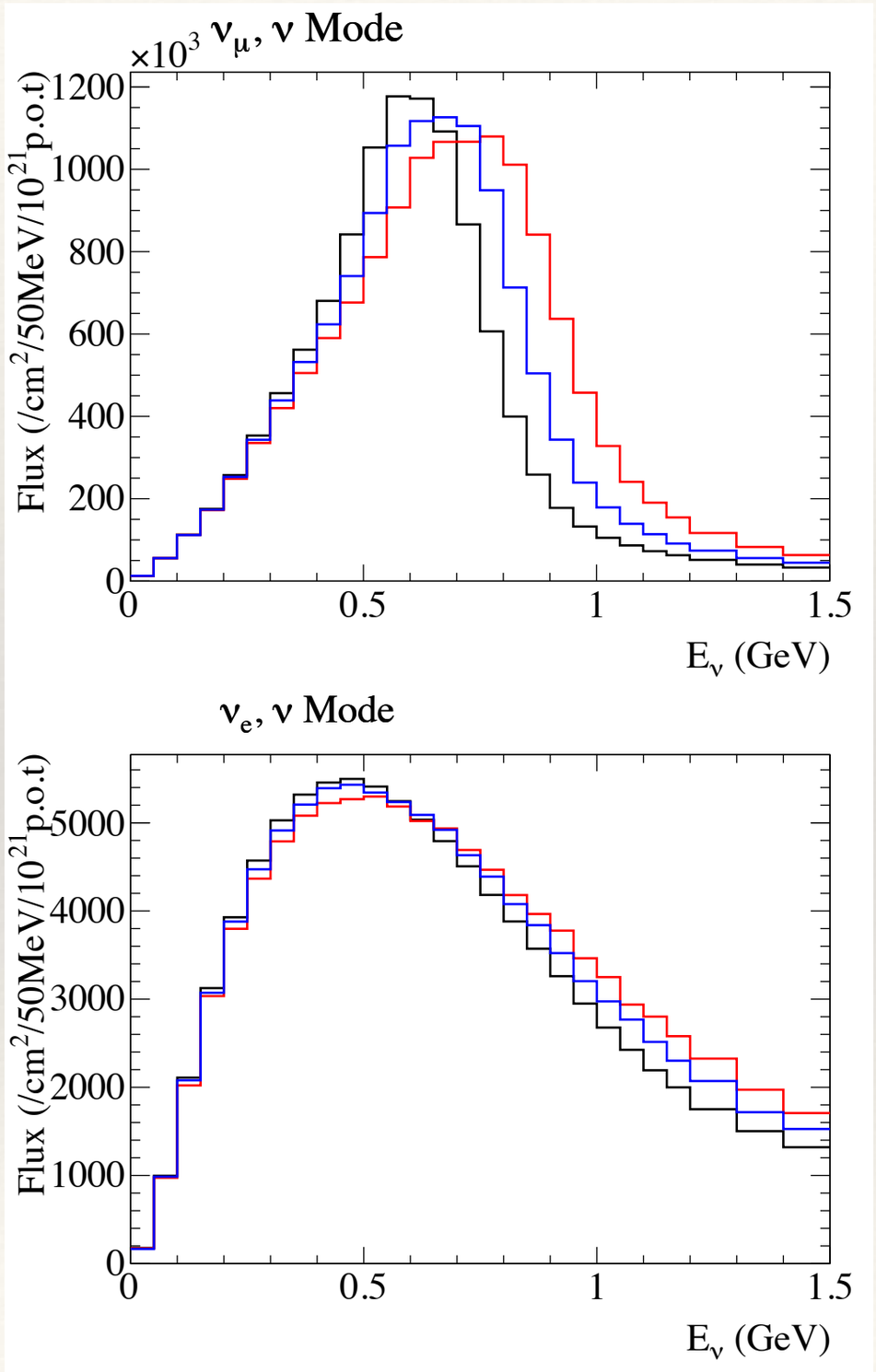
Oscillation Probability (IH)

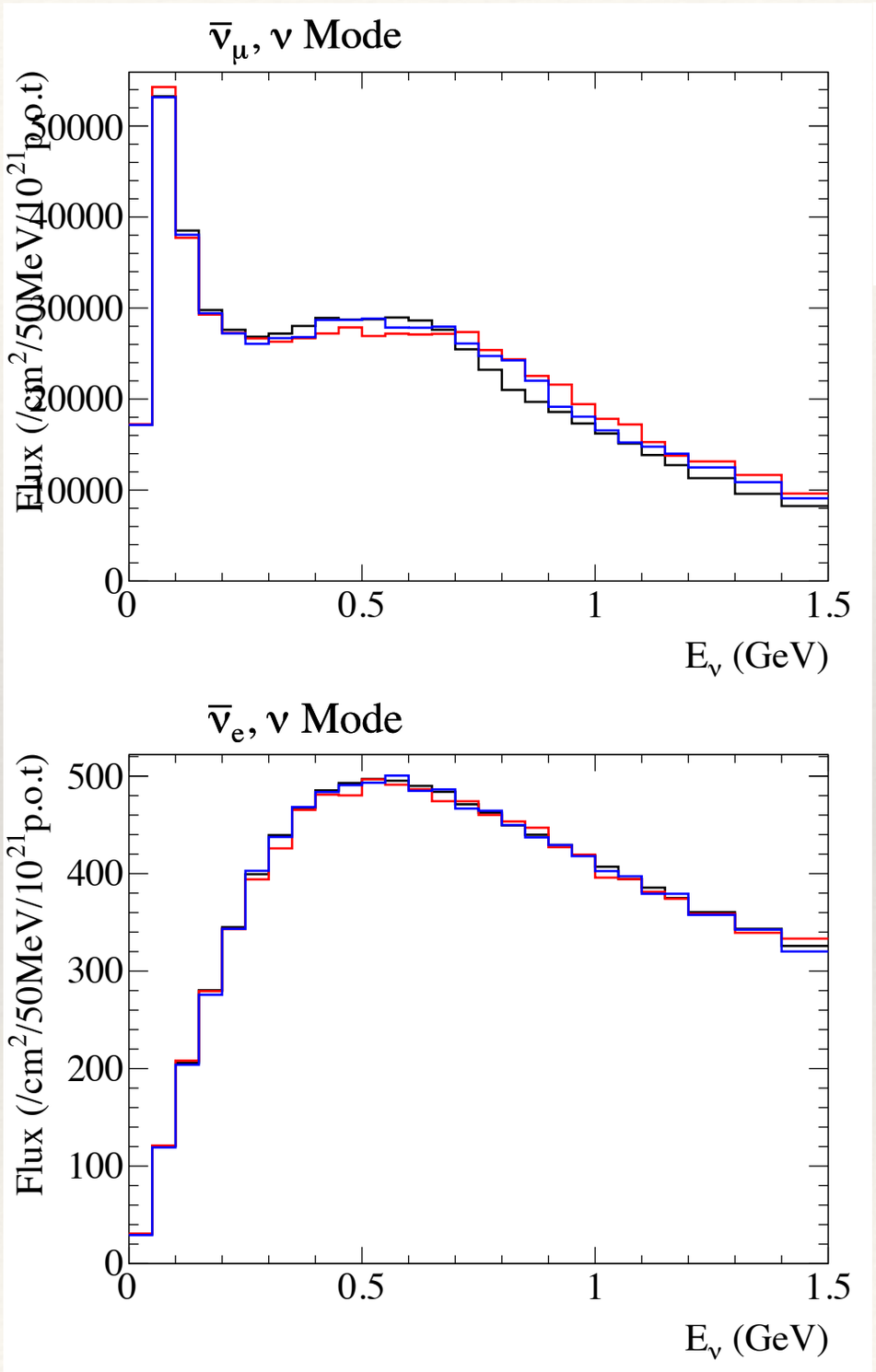




- For inverted hierarchy, the peak energies are similar
- May get improved statistical sensitivity by shifting the neutrino spectrum higher in energy

Neutrino Mode Fluxes



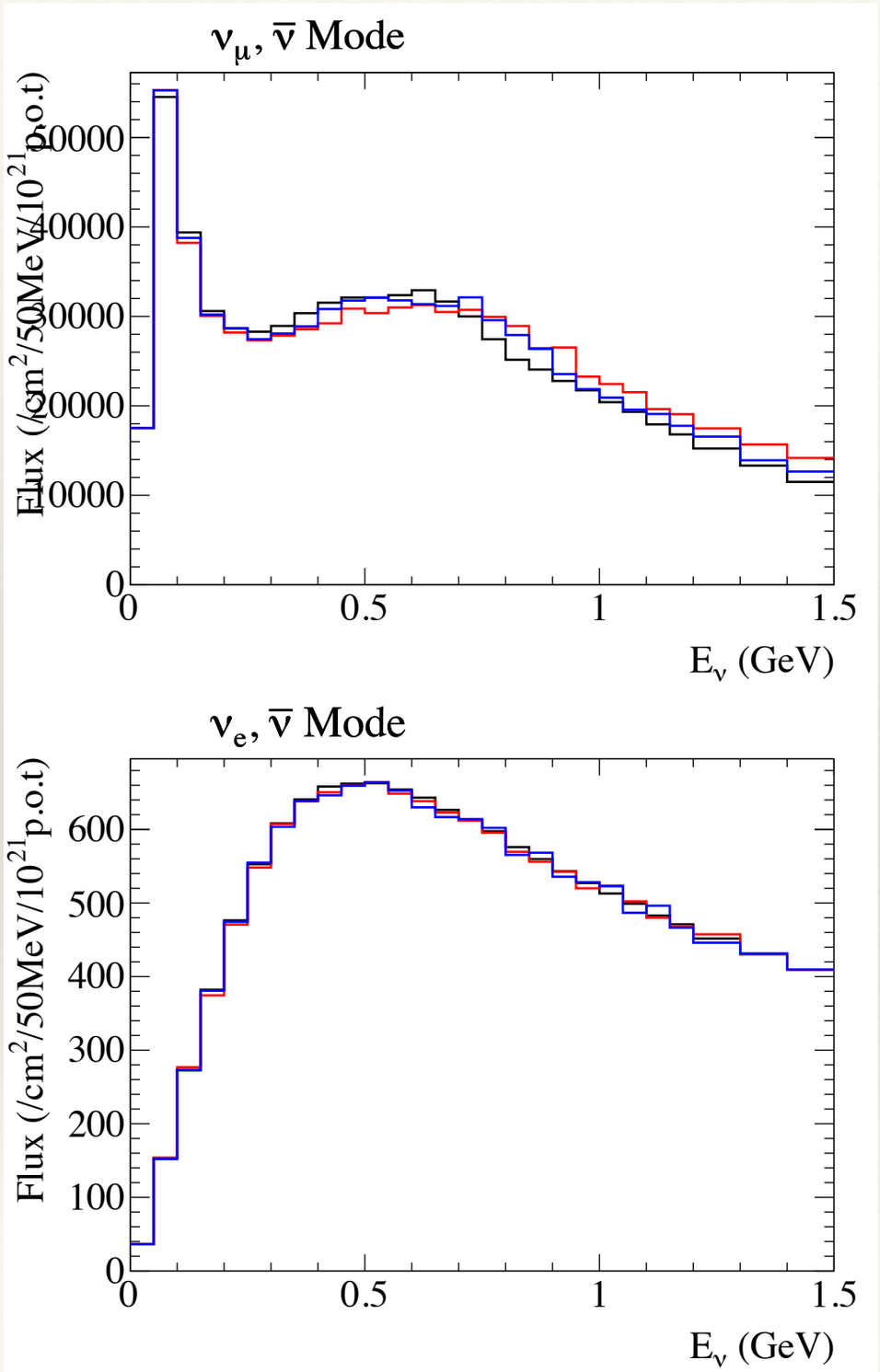


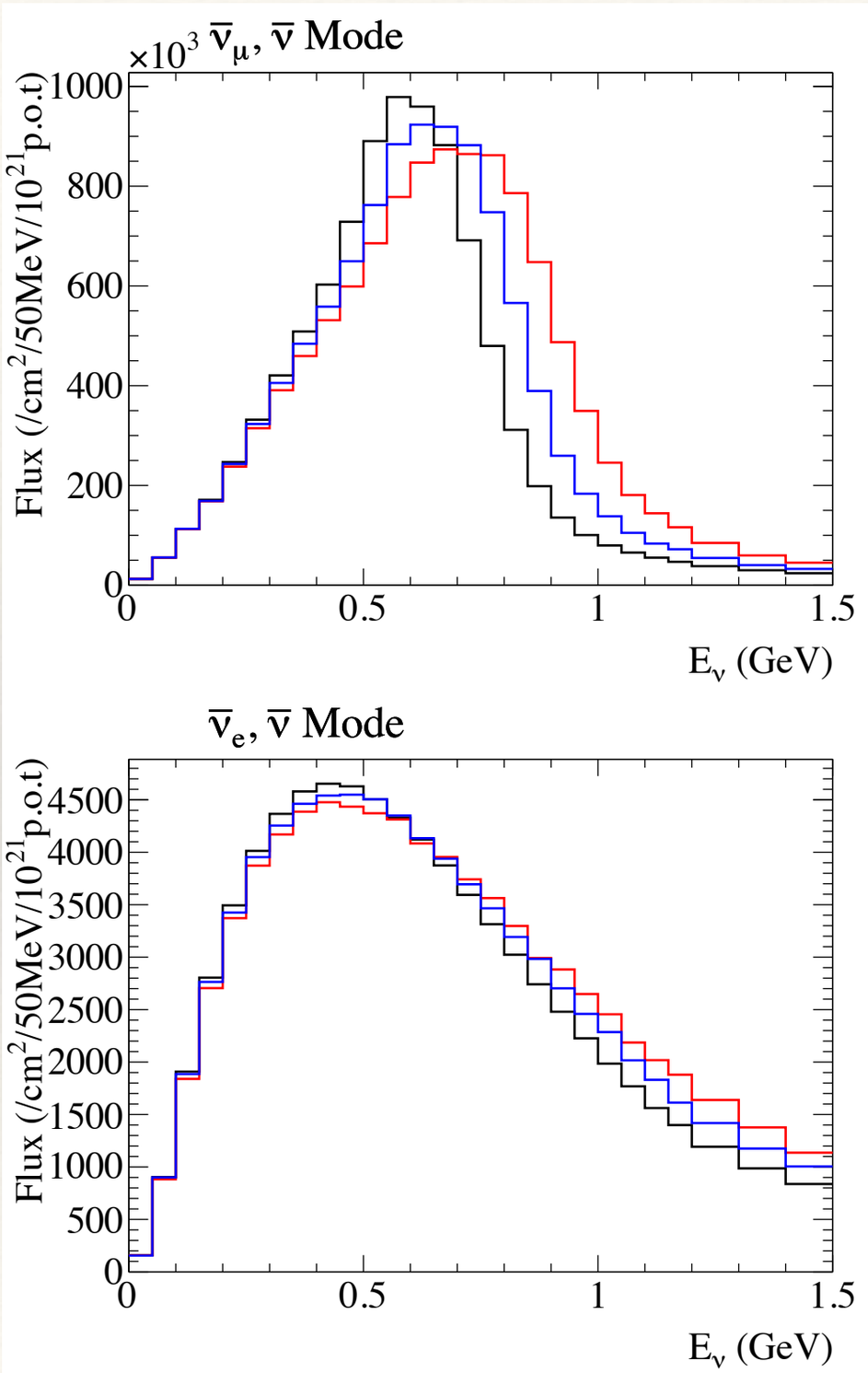
θ_{OA}=2.0°

θ_{OA}=2.25°

θ_{OA}=2.5°

Anti-Neutrino Mode Fluxes





θ_{OA}=2.0°

θ_{OA}=2.25°

θ_{OA}=2.5°

Comments on Fluxes

- Decreasing the off-axis angle increases the peak energy and high energy tail for the right-sign muon neutrino flux, as expected
- Essentially no effect on the wrong-sign fluxes
- A small shift to higher energy in the right-sign electron neutrino flux
- Have produce ratio histograms to reweight the 2.5 degree flux to 2.0 or 2.25 degrees:

http://www.t2k.org/asg/oagroup/tmp/fluxoadependence/at_download/file

- File contains histograms with energy dependent weights to transform from the nominal flux to 2.0 or 2.25 degrees off-axis

Histogram names: weights_oa_<angle>_<horn mode>_<flavor>

2p0 (2.0) or 2p25 (2.25)

numode (neutrino mode) or anumode (antineutrino mode)

numu, numub, nue or neub

Conclusion

- Considering the large value for θ_{13}, the reduced NCπ^0 background and the plan for significant antineutrino mode data, a flux peaked at 600 MeV may not be optimal
- In principle, the T2K beam line can be reconfigured to off-axis angles as small as 2.0 degrees (but it would be significant work)
- It is interesting to see how the sensitivity is affected by changing the off-axis angle
- Have produced flux weights to reweight to 2.0 or 2.25 degree off-axis angles
- Can sensitivities from ultimate T2K statistics be checked? For now I think it is ok to treat systematic errors and the ND constraint as unchanged for the off-axis angle changes.